Zeeman effect

Hydrogen atom energy levels in a magnetic field $B = |\mathbf{B}|$ are

$$E_{nljm_j} = -\frac{\alpha^2\mu c^2}{2n^2}\left[1 + \frac{\alpha^2}{n^2}\left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right)\right] + g_J m_j \mu_B B$$

where

$$l = 0, 1, 2, \ldots, n-1$$

and

$$j = \left|l \pm \tfrac{1}{2}\right|, \quad m_j = \pm\tfrac{1}{2}, \pm\tfrac{3}{2}, \ldots, \pm j$$

Symbol g_J is the Landé g-factor

$$g_J = 1 + \frac{j(j+1) - l(l+1) + \frac{3}{4}}{2j(j+1)}$$

For $n = 2$ there are eight energy levels.

n	l	j	m_j	E_{nljm_j}
2	1	$\frac{3}{2}$	$\frac{3}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{1}{16}\alpha^2) + 2\mu_B B$
2	1	$\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{1}{16}\alpha^2) - 2\mu_B B$
2	1	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{1}{16}\alpha^2) + \frac{2}{3}\mu_B B$
2	1	$\frac{3}{2}$	$-\frac{1}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{1}{16}\alpha^2) - \frac{2}{3}\mu_B B$
2	1	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{5}{16}\alpha^2) + \frac{1}{3}\mu_B B$
2	1	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{5}{16}\alpha^2) - \frac{1}{3}\mu_B B$
2	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{5}{16}\alpha^2) + \mu_B B$
2	0	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{\alpha^2\mu c^2}{8}(1 + \frac{5}{16}\alpha^2) - \mu_B B$

Eigenmath script